

What's The Rate?

How many jumping jacks can you do in 10 seconds? $\frac{\text{jumping jacks}}{\text{seconds}}$

Predict how many you could do in 30 seconds. _____ = _____

How many snaps can you do in 10 seconds? $\frac{\text{snaps}}{\text{seconds}}$

Create a unit rate of snaps per second. _____ = _____

Definition	Examples
Ratio: A multiplicative comparison of two <u>quantities</u> .	2 cats to <u>3 dogs</u> , 2 cats : <u>3 dogs</u> 2 cats out of <u>5 animals</u> , $\frac{2 \text{ cats}}{3 \text{ dogs}}$
Rate: A comparison by <u>division</u> of two quantities having different <u>attributes</u> .	<u>\$10 for 2 pounds</u> <u>150 miles in 5 hours</u>
Unit Rate: A <u>rate</u> that is <u>simplified</u> so that it has a denominator of <u>1</u> .	<u>\$5 for 1 pound</u> <u>30 miles per hour</u>

Determine the rate for each situation.

1. There are 117 players on 9 teams. How many players on one team?

$9 \overline{)117}$
 $\underline{-9}$
 27

$117 \text{ players} \xrightarrow{\div 9} x$
 $9 \text{ teams} \xrightarrow{\div 9} 1 \text{ team}$

$x = 13 \text{ players}$

2. There are 56 questions on 4 quizzes. How many questions are on one quiz?

$56 \text{ Qs} \xrightarrow{\div 4} x$
 $4 \text{ quiz} \xrightarrow{\div 4} 1 \text{ quiz}$

$x = 14 \text{ Qs}$

3. It takes you 60 minutes to read 15 pages. How long will it take you to read one page?

$60 \text{ min} \xrightarrow{\div 15} x$
 $15 \text{ pg} \xrightarrow{\div 15} 1 \text{ pg}$

$x = 4 \text{ min}$

4. Use the given rate to solve the problem and determine an expression that could be used to solve any problem involving the given rate. If the perimeter of a square is 20 inches with a side length of 5 inches, what is the perimeter of a square with a side length of 10?

5 in. side

Side Length	Rule	Perimeter
5	$4(5)$	20
6	$4(6)$	24
7	$4(7)$	28
10	$4(10)$	40
s	$4(s)$	p

What is the perimeter of a square with a side length of 10 inches?

$\times 4$ $\frac{5 \text{ in. side}}{20 \text{ in. perimeter}} = \frac{10 \text{ in. side}}{40 \text{ perimeter}}$ $\times 4$

5. Use the given rate to solve the problem and determine an expression that could be used to solve any problem involving the given rate. If the rate is 365 days in one year, how many days are in 4 years?

Years	Rule	Days
1	365(1)	365
2	365(2)	730
4	365(4)	1460
y	365(y) =	d

How many days are in 4 years?

$$\frac{365 \text{ Days}}{1 \text{ year}} = \frac{1460 \text{ Days}}{4 \text{ years}}$$

$$\begin{array}{r} 365 \\ \times 4 \\ \hline 1460 \end{array}$$

For problems 6 – 9, find the **cost** for each situation.

6. If you pay \$36 for 4 DVDs, how much will you pay for one DVD?

$$\frac{\$36}{4 \text{ DVD}} = \frac{\$9}{1 \text{ DVD}}$$

8. If you pay \$15 for 3 ounces, how much will you pay for one ounce?

$$\frac{\$15}{3 \text{ oz}} = \frac{\$5}{1 \text{ oz}}$$

7. If you pay \$65 for 5 books, how much will you pay for one book?

$$\frac{\$65}{5 \text{ book}} = \frac{\$13}{1 \text{ book}}$$

9. If you pay \$5 for 10 pens, how much will you pay for one pen?

$$\frac{\$5}{10 \text{ pens}} = \frac{\$0.50}{1 \text{ pen}}$$

Determine the rate for each given situation. Use these rates to respond to part c) in each problem.

10. Tom jogs 4 miles in 36 minutes. Beau jogs 2 miles in 24 minutes.

- a) Determine how long it takes Tom to jog 1 mile.

$$\frac{36 \text{ minutes}}{4 \text{ miles}} = \frac{9 \text{ min}}{1 \text{ mile}}$$

- b) Determine how long it takes Beau to jog 1 mile.

$$\frac{24 \text{ min}}{2 \text{ mi}} = \frac{12 \text{ min}}{1 \text{ mi}}$$

- c) Who jogs faster? How many minutes per mile faster?

Tom is faster by 3 minutes per mile.